NOVAMED SALES & CUSTOMER ANALYSIS

Data Analytics Capstone Project

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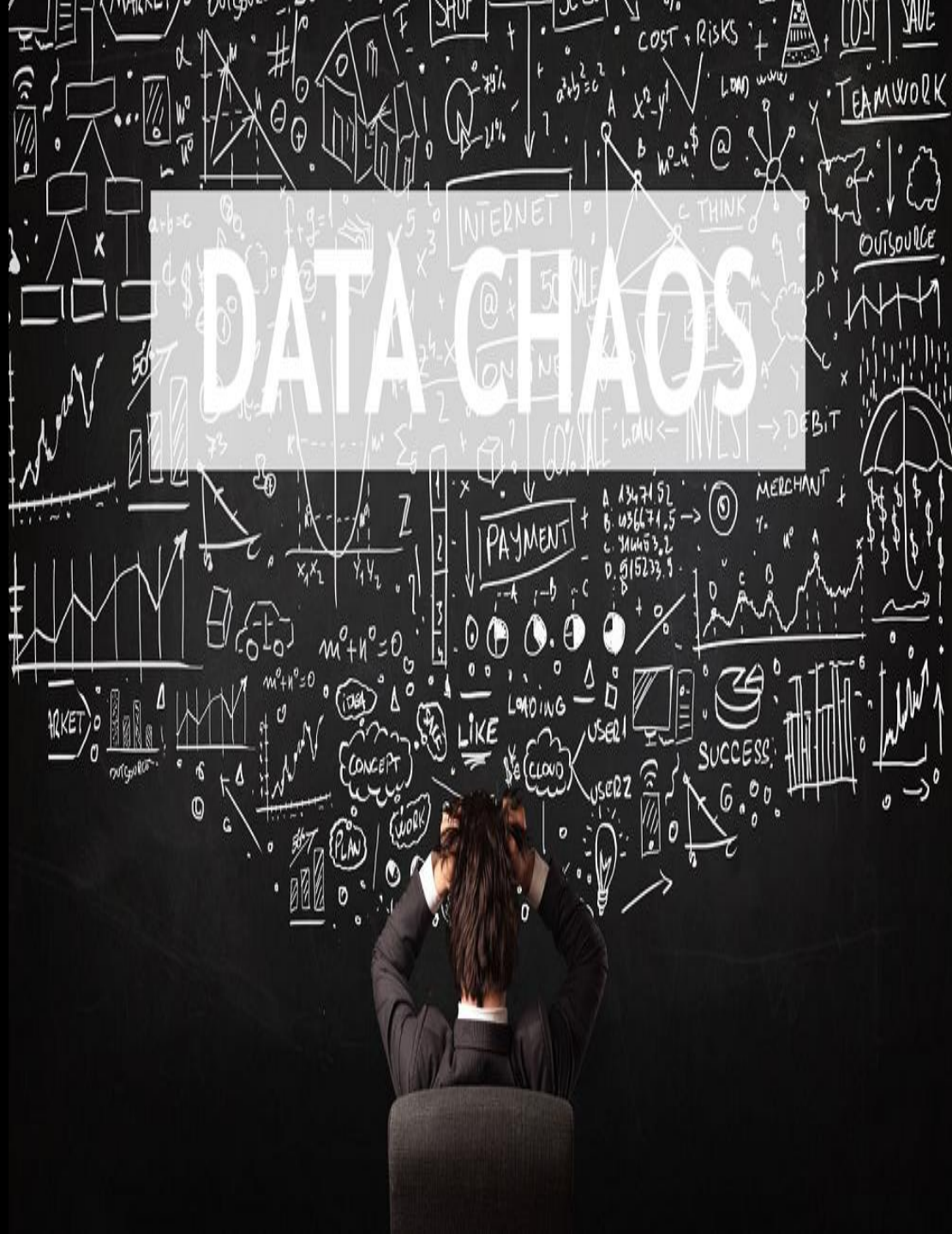
BUSINESS OVERVIEW

- NovaMed is a pharmaceutical company specializing in drug production and distribution
- The company operates across multiple countries
- It offers a wide range of medical products
- The business focuses on improving healthcare through accessible medication

BUSINESS PROBLEM

- NovaMed lacks clear visibility into sales performance
- The company is unable to identify top-performing products
- Customer purchasing behavior is not fully understood
- Decision-making is not fully data-driven



PROJECT OVERVIEW

- Analyze NovaMed’s 2023 sales performance
- Identify key revenue drivers and trends
- Understand customer purchasing behavior
- Support data-driven decision-making

DATASET STRUCTURE

FactTable

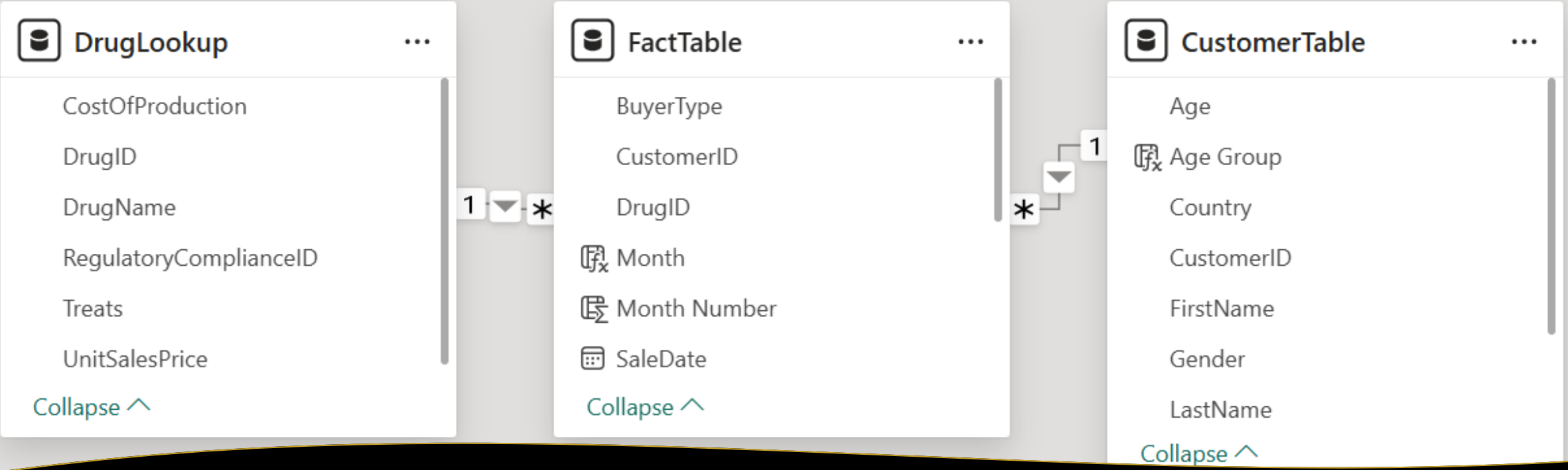
- Sales transactions
- Units sold
- Sale date
- Buyer type

CustomerTable

- Customer demographics
- Age & Gender
- Country

DrugLookup

- Drug name
- Unit price
- Cost of production



DATA CLEANING & MODELING

- Cleaned and validated dataset for consistency
- Ensured correct data types (dates, numbers, categories)
- Created relationships between tables using:
  - **CustomerID**
  - **DrugID**
- Structured data model for accurate analysis

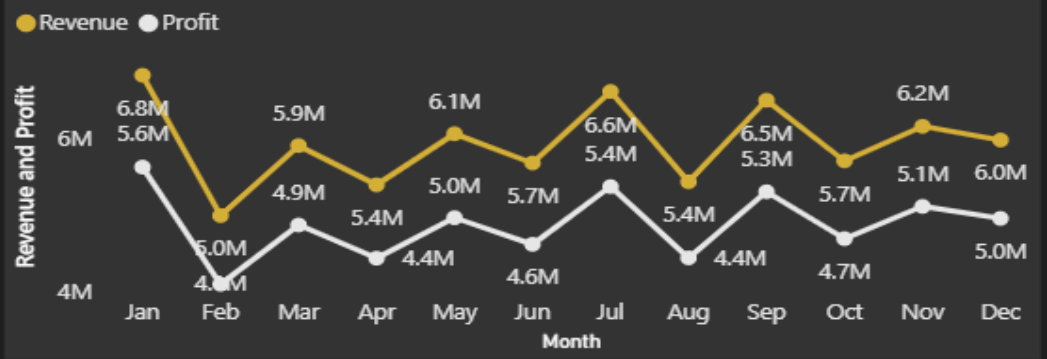
SALES PERFORMANCE DASHBOARD

NovaMed Sales Performance

Revenue, Profit & Product Performance Overview (2023)

71.31M	58.45M	12.85M	82%
Total Revenue	Profit	COGS	Profit Margin

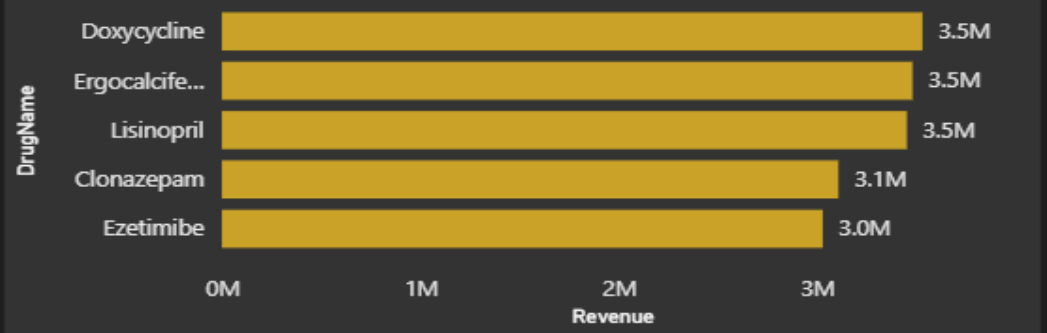
Revenue and Profit by Month



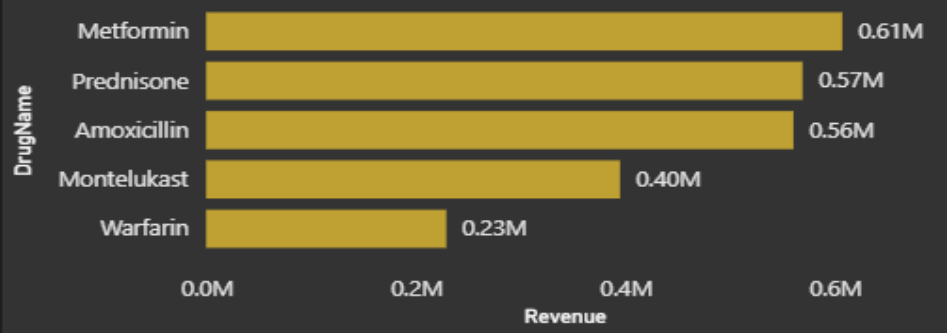
Top 5 Customers by Revenue

CustomerID	Revenue
191	803,622.31
115	624,971.63
150	606,665.90
196	603,145.93
81	592,823.45
Total	**3,231,229.22**

Top 5 Drugs by Revenue



Bottom 5 Drugs by Revenue



BuyerType

- Seller
- User

Country

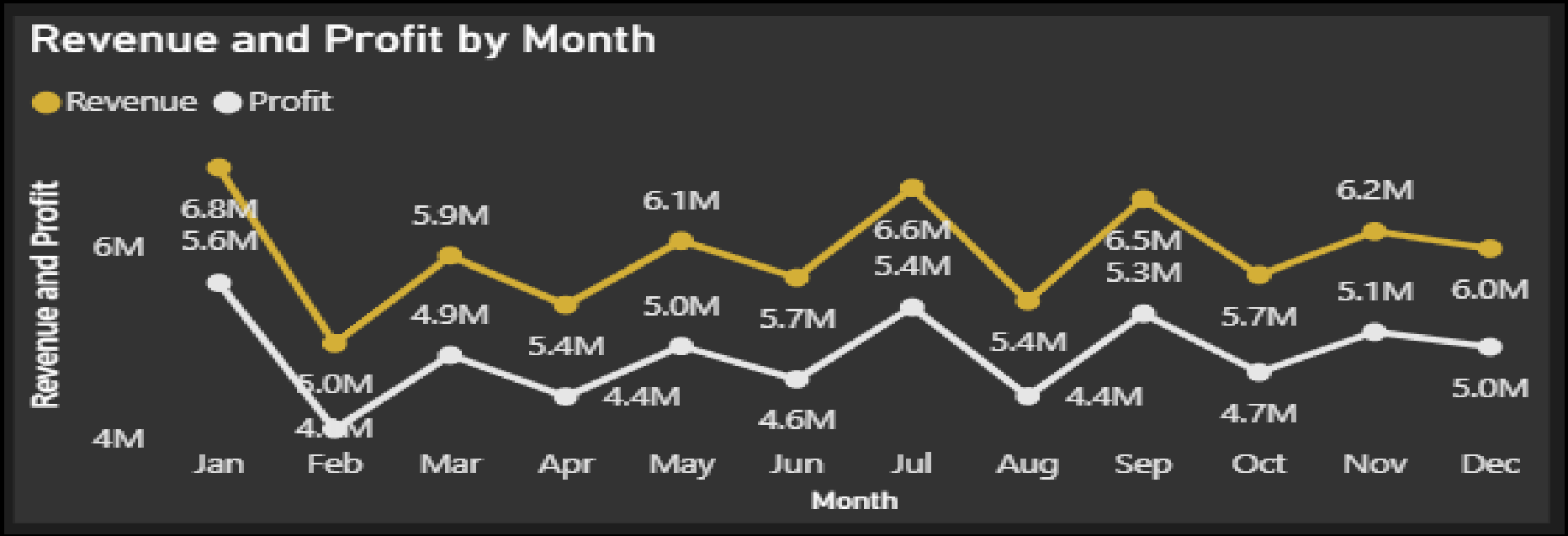
- Australia
- Germany
- Canada
- United Kingdom
- France
- United States

DrugName

- Albuterol
- Amlodipine
- Amoxicillin
- Amphetamine

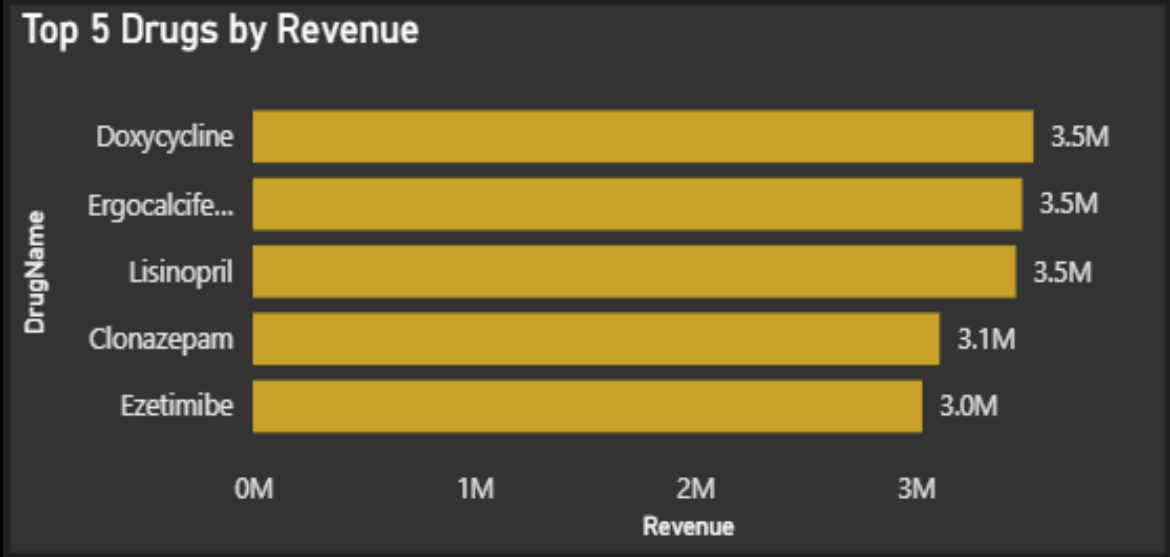
Overview of revenue, profit, and product performance

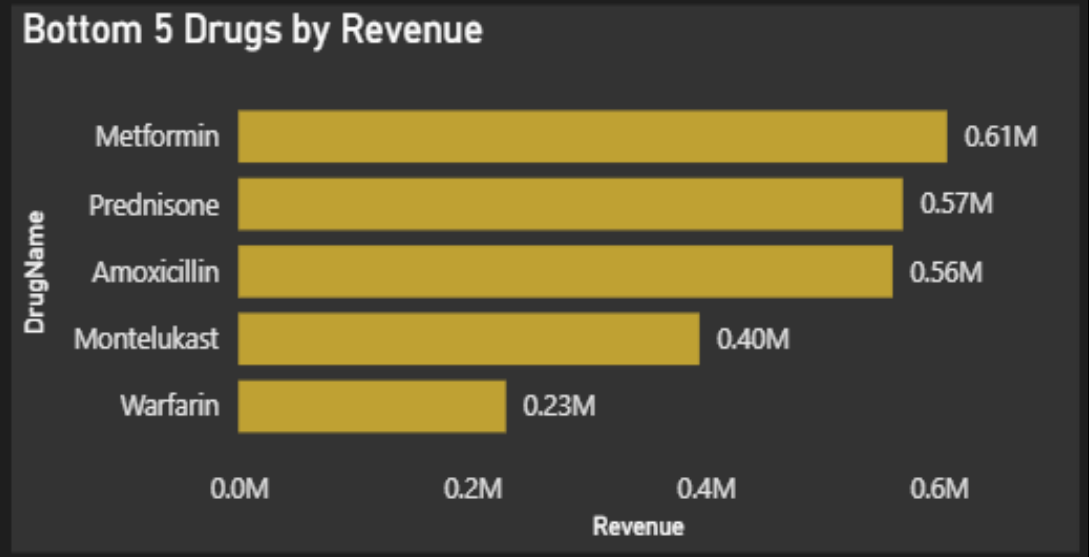
REVENUE & PROFIT TREND ANALYSIS



- Revenue and profit fluctuate across different months
- Trends indicate variations in sales performance over time
- Certain periods show stronger performance than others
- Highlights the need for strategies to stabilize revenue

PRODUCT PERFORMANCE ANALYSIS





- A few drugs contribute significantly to total revenue
- Top-performing drugs drive the majority of sales
- Some drugs consistently underperform
- Opportunity exists to improve or reposition low-performing products

CUSTOMER REVENUE CONTRIBUTION

Top 5 Customers by Revenue

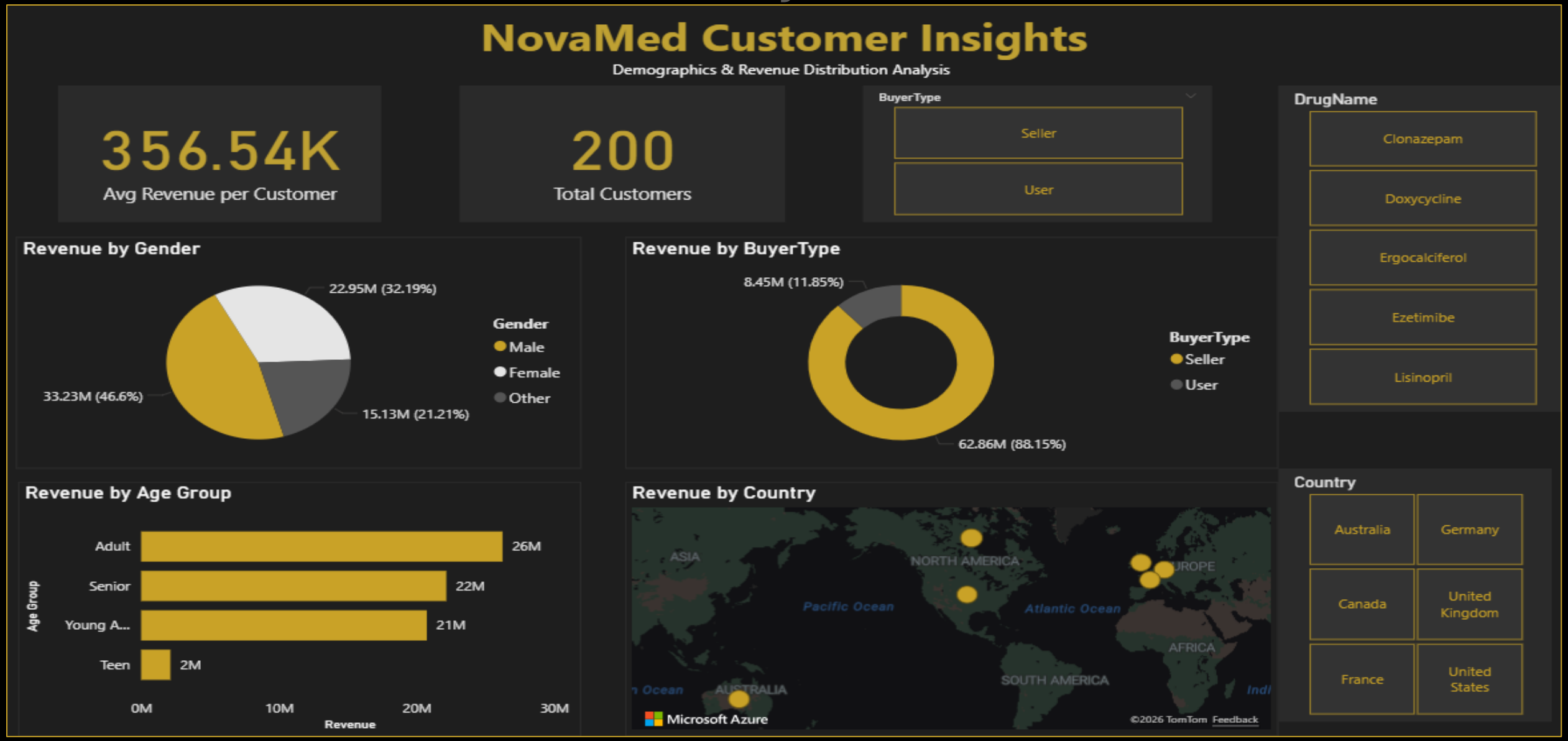
CustomerID	Revenue
196	603,145.93
191	803,622.31
150	606,665.90
115	624,971.63
81	592,823.45
Total	**3,231,229.22**

- Revenue is concentrated among a few key customers
- Top customers contribute a significant portion of total sales
- Indicates the importance of high-value customer segments
- Opportunity to strengthen relationships with key customers

SALES PERFORMANCE RECOMMENDATIONS

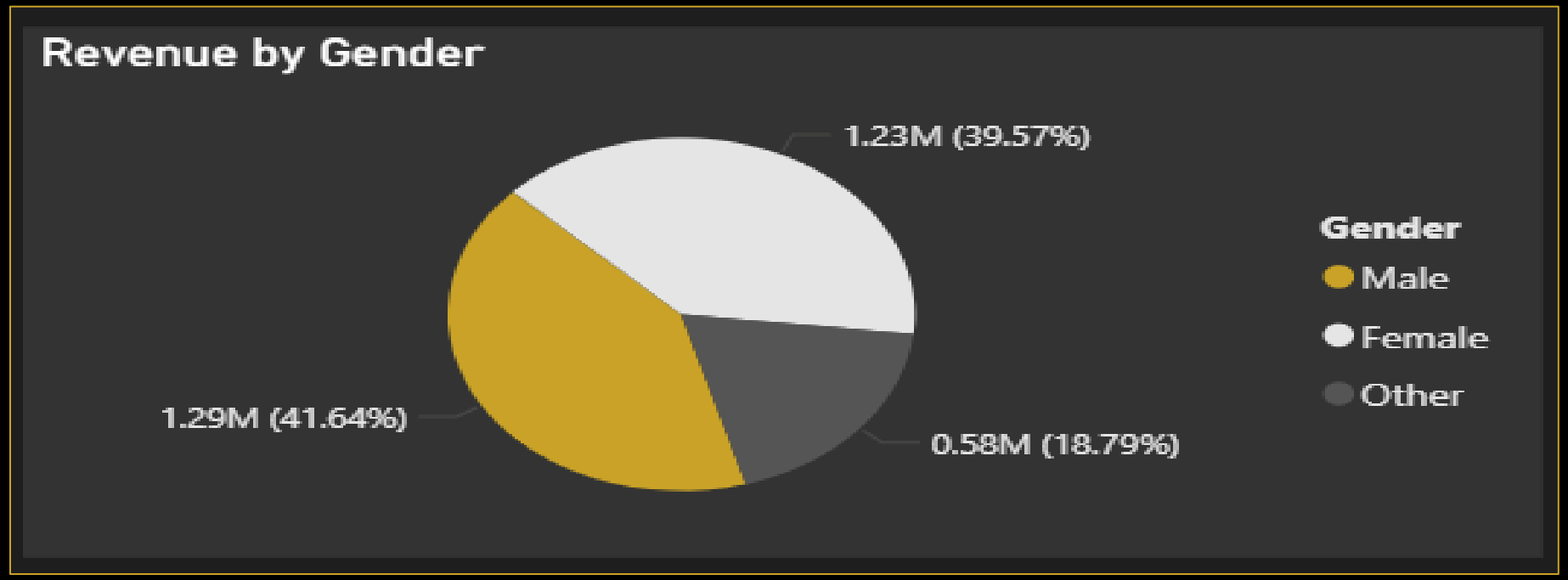
- Prioritize production and distribution of top-performing drugs
- Reduce losses from underperforming products through review or repositioning
- Implement demand forecasting to stabilize monthly revenue fluctuations
- Optimize pricing and cost structure to improve profit margins
- Strengthen supply chain efficiency to ensure product availability

Customer Analysis Dashboard



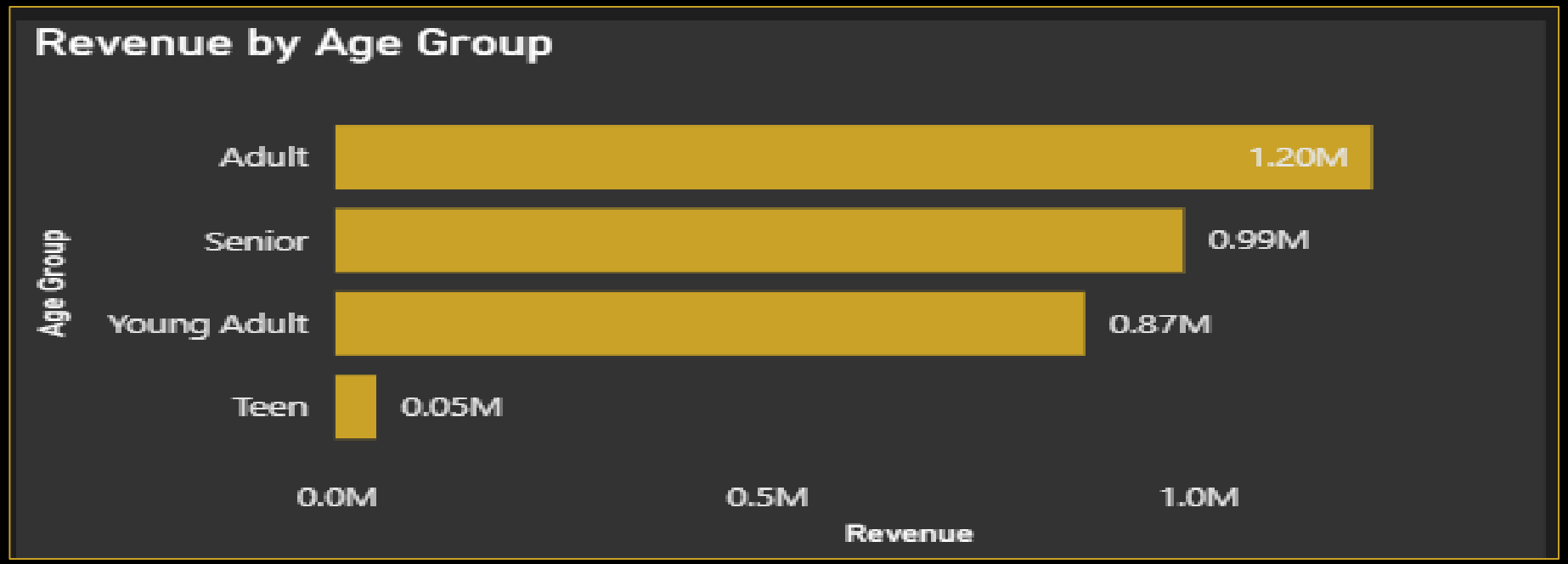
This dashboard presents key insights into customer demographics, buying behavior, and revenue contribution across different segments and regions.

REVENUE DISTRIBUTION BY GENDER



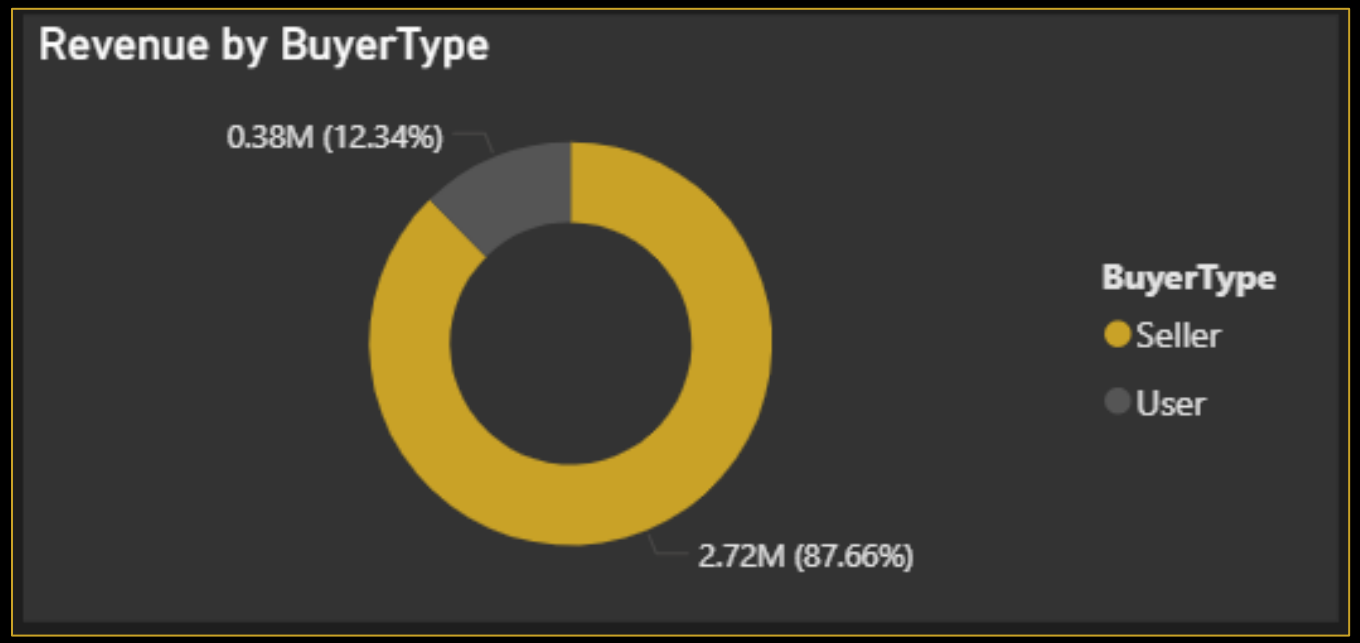
- Revenue is distributed across different gender groups
- Male customers contribute the highest share of revenue
- Female customers also represent a significant portion
- Indicates variation in purchasing behavior across gender segments

CUSTOMER DEMOGRAPHIC INSIGHTS



- Revenue varies across age groups and gender
- Adult customers generate the highest revenue
- Male customers contribute slightly more than others
- Different demographic groups show different purchasing patterns

BUYER TYPE ANALYSIS



- Sellers contribute the majority of revenue
- Users contribute a smaller share
- Indicates strong influence of bulk or business buyers
- Shows variation in purchasing behavior

GEOGRAPHIC SALES INSIGHT

- Revenue is distributed across multiple countries
- Some regions contribute more significantly than others
- Indicates presence of strong markets
- Highlights opportunities for expansion

CUSTOMER INSIGHTS RECOMMENDATIONS

- Focus on high-value customer segments (Adults & bulk buyers)
- Develop targeted marketing campaigns based on gender and demographics
- Introduce loyalty and retention programs to increase customer lifetime value
- Provide incentives for bulk buyers (sellers) to drive higher sales volume
- Expand presence in high-performing countries while growing emerging markets

CONCLUSION

- NovaMed demonstrates strong overall sales performance, driven by a few high-performing products
- Revenue fluctuations highlight the need for better demand planning and sales stability
- Customer analysis shows that adults and bulk buyers contribute the highest value
- Sales performance varies across regions, indicating key markets and growth opportunities
- Data-driven insights provide a foundation for improving product strategy, customer targeting, and business decisions

THANK YOU